

KRISHI-AGENT

Hierarchical Crop Disease Classification and Conflict-Aware
Multi-Agent System for Telugu Farm Advisory

Full Technical Evaluation Report

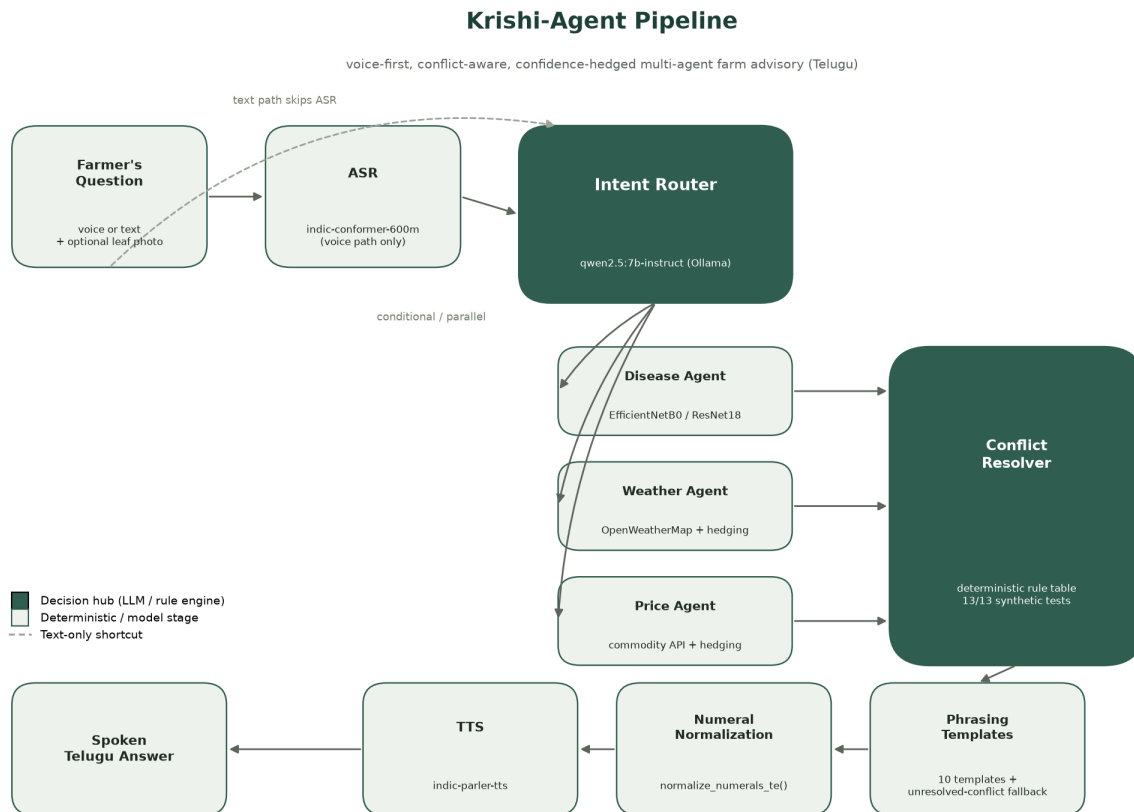
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Project Guide	Dr. Nagaraju
Department	[not stated in project source files -- fill in before submission]
Compiled	September 2026
Repository	harikharreddy/Krishi-Agent

Every number in this report traces to a script, a dataset, or a dated evidence file in the repository -- nothing is estimated. Diagrams are labeled as either reused directly from tonight's evidence files, or newly generated from that same underlying real data where no image previously existed.

System Architecture

A farmer asks a question -- spoken or typed, in Telugu -- optionally attaching a leaf photo, and receives one spoken Telugu answer combining disease diagnosis, weather, and market price, with contradicting advice explicitly resolved rather than silently dropped.

Diagram newly generated (docs/report_assets/generate_architecture_diagram.py) -- no architecture diagram image existed anywhere in the repository. Stages and order are taken directly from evaluation_and_validation.md Sec. 1 and the real call order in backend/voice.py (normalize_numerals_te() is a real, separate function call before TTS -- shared/text_normalization.py).



Executive Summary

Krishi-Agent substantiates **three system-level contributions**, not a single component-level accuracy claim: (1) explicit, independently-tested conflict arbitration across competing advice signals; (2) confidence calibration validated empirically via a five-signal convergence across four independently-trained checkpoints spanning two distinct architecture families, plus a 5-fold cross-validation variance estimate; (3) a confirmed, cross-speaker structural failure mode at the ASR-to-intent-router boundary.

Headline findings

Metric	Result	Notes
Deterministic logic test suite	71/71 (100%)	Conflict resolution, phrasing, state-mapping, edge cases
Intent Router accuracy	81.8% held-out (87.5% full set)	87.5% includes 5 few-shot-exposed questions -- 81.8% is the honest estimate
Disease Agent, real-world (zero-shot)	21.45-23.45% across 4 checkpoints, 2 architectures	Converges tightly; 5-fold CV: 23.38% +/- 0.83%
Disease Agent, real-world (deployed, fine-tuned)	37.65% (n=85, test-only)	Exploratory result -- not a like-for-like comparison to the zero-shot rows (11x smaller n)
ASR word error rate	11.19% (n=64, 4 speakers)	Range 8.09-13.62% across speakers
Voice-vs-text ablation	58/64 exact match	All 6 mismatches explained -- 4 structural (ASR pH transliteration), 2 live-API timing
Conflict Resolver	13/13 synthetic tests (7/36 of full 3-agent state grid)	1 real gap found via live-data testing, fixed
Cross-architecture convergence (McNemar's)	6/6 pairwise comparisons non-significant	p=0.316/0.754/0.179 (same-family) + 0.369/0.156/0.083 (cross-architecture)

Novelty Claims and Closest Existing Work

Framed at the **system level**, not as a component-level accuracy claim. The Disease Agent's own ~22% real-world zero-shot accuracy is reported as a measured, structurally-bounded limitation (Sec. 5), not positioned as the project's contribution.

1. Conflict arbitration between competing advice signals

Explicitly engineered and independently tested -- a deterministic rule table (13/13 on synthetic tests, 7/36 of the full 3-agent-active state grid), one real gap found via live-data testing and fixed. Not described as present in any of the three comparable systems surveyed.

2. Confidence calibration validated empirically, not merely asserted

A five-signal convergence (accuracy band, formal confusion matrix, Expected Calibration Error, a formal McNemar's-test significance check, and a fourth independently-architected checkpoint reproducing all of the above) across four checkpoints spanning two architecture families -- plus a 5-fold cross-validation showing the estimate itself is stable (23.38% +/- 0.83%) under resampling of the training data.

3. A confirmed, cross-speaker structural failure mode at the ASR-to-intent-router boundary

A domain-specific technical term (soil pH) transliterated by ASR flips the intent router's classification, reproduced independently across all 4 recorded speakers -- not a single-speaker artifact.

Closest existing work

Three systems read in enough depth to compare honestly, all pulled directly from krishiagent-literaturesurvey.docx (our own 25-paper survey) -- Farmer.Chat [15] (arXiv:2409.08916), Kisaan Margadarshak [13] (Trimukhe et al. 2025, Springer, DOI:10.1007/978-3-031-74440-2_1, rated HIGH in our survey and already the base paper the rest of this report treats as the closest structural precedent), and CropCare Companion [14] (Sable et al. 2025, IJRASET, rated CAUTION in our survey).

Dimension	Krishi-Agent	Farmer.Chat [15]	Kisaan Margadarshak [13]	CropCare Companion [14]
Architecture	Multi-agent + explicit rule-based Conflict Resolver	RAG + multi-agent orchestration	Android app -- on-device MobileNetV2 CNN + Django + Flutter, weather + mandi price APIs	Naive Bayes + DeepSeek LLM fallback; NLP pipeline + Google Translate + Web Speech API
Explicit conflict resolution	Yes -- 13/13 synthetic tests (7/36 of full 3-agent state grid)	Not described	Not described in survey entry	Not described in survey entry
Disease diagnosis	Own trained CV model, in-pipeline (37.65% honestly measured)	Delegated to Plantix (3rd party)	On-device CNN, single-crop (cotton) only -- no quantified accuracy/F1	No image-based detection -- named as future work
Confidence shown to user	Yes -- hedged phrasing + UI indicator	Not described (retrospective thumbs-up/down only)	Not described in survey entry	Not described in survey entry
Evaluation scale	Component-level (n=12-263)	15,000+ users, 300,000+ queries	No accuracy/F1 or user-scale reported (survey's own stated limitation)	>91% accuracy, multilingual -- but proprietary, undisclosed-size dataset (CAUTION-rated)

Farmer.Chat massively outpaces this project on real-world deployment and user-study rigor -- a genuine, uncontested strength of theirs. CropCare Companion's own reported >91% accuracy is also higher than any Krishi-Agent number in this report, stated here rather than omitted -- but it's measured on a proprietary, undisclosed-size Q&A dataset with no independent evaluation, exactly why our own survey rated that paper CAUTION rather than HIGH, and why it isn't treated as a like-for-like comparison against this report's own sourced-and-sized accuracy figures. This table supports a narrower, verifiable, system-level claim, not a claim of being "better overall."

Metric 1 -- Disease Agent Real-World Accuracy

The most extensively evaluated component. Baseline: PlantVillage (studio photos, training distribution) 99.7% vs. PlantDoc (real-world field photos, zero-shot) 28.2% on the original 85-image test-only baseline -- a 71.5-point domain-shift drop, with a specific attractor-class failure mode (Tomato_Late_blight over-predicted).

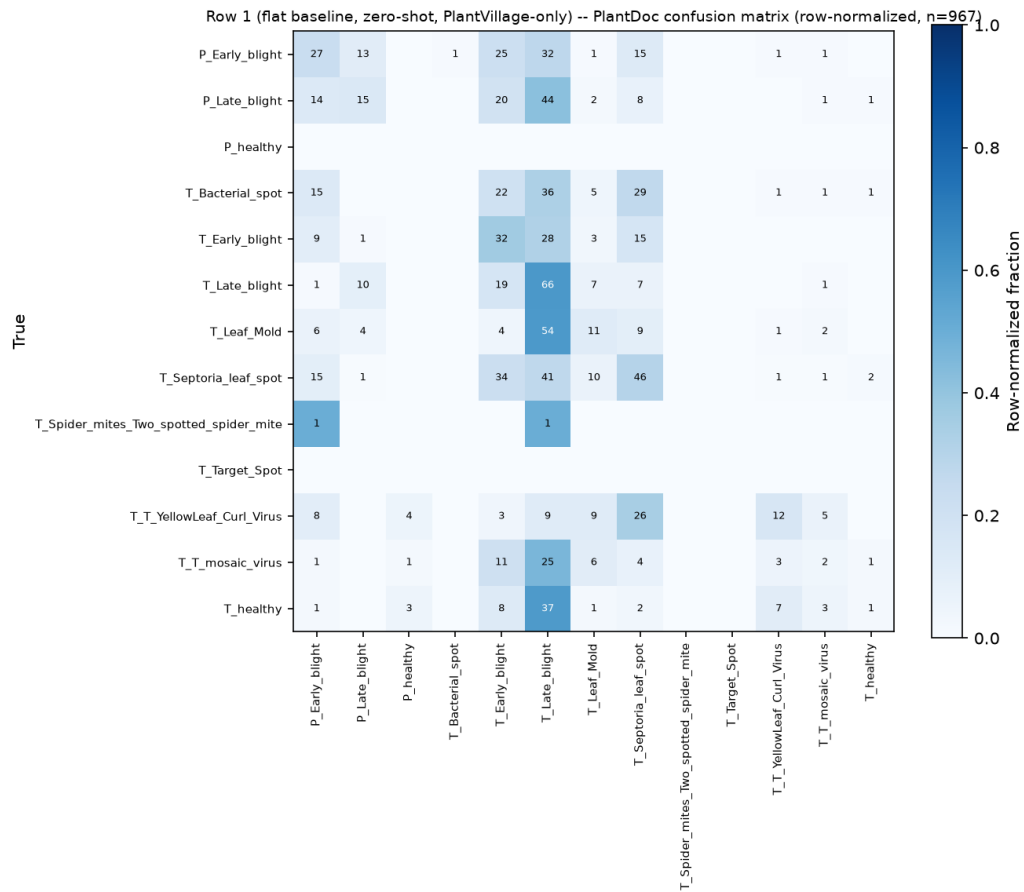
Full model lineage (all 8 rows)

#	Checkpoint	Trained on	PlantDoc accuracy	Deployed?	Footprint
1	flat_baseline_best.pt	PlantVillage only (flat)	22.00% (n=968)	No	4.02M params, 0.82 GFLOPs
2	stage1/2_*_plantvillage_only.pt	PlantVillage only (hierarchical)	23.45% (n=968) / 28.24% (n=85)	No	12.04M params, 1.64 GFLOPs/inf
3	stage1/2_*_plantdoc_no_target_spot.pt	+ PlantDoc train split	40.0% (n=85, test-only)	No (superseded)	same as row 2/5
4	stage1/2_*_pre_potato_healthy.pt	+20 external Target_Spot photos	38.8% (n=85, test-only)	No (superseded)	same as row 2/5
5	stage1/2_*_best.pt	+100 external Potato_healthy photos	37.65% (n=85, test-only)	YES (deployed)	12.04M, 240.2ms latency, 4.16 img/s
6	disease_agent_efficientnetb0.pt (v2)	PlantVillage only, indep. Colab run	21.45% (n=965)	No	4.02M params, 0.82 GFLOPs
7	disease_agent_plantdoc_finetuned_best.pt	Row 6 + PlantDoc train, deduped	INVALID -- bug-confirmed (n=71, 2 classes silently dropped)	No	same as row 6
8	disease_agent_resnet18_baseline.pt	PlantVillage only (ResNet18)	22.51% (n=822) / 22.41-22.47% apples-to-apples	No	11.18M params, 3.65 GFLOPs

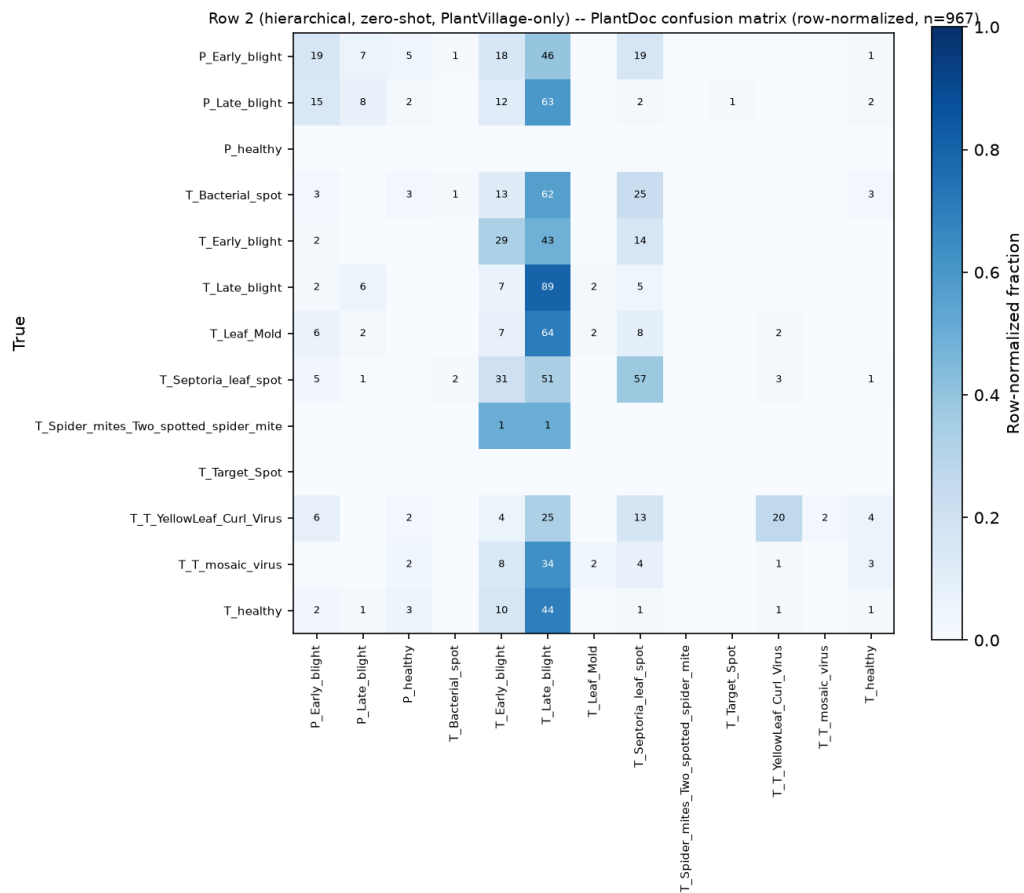
Reused from docs/evidence/model_lineage.md's lineage table.

Confusion matrices, rows 1, 2, 6, 8

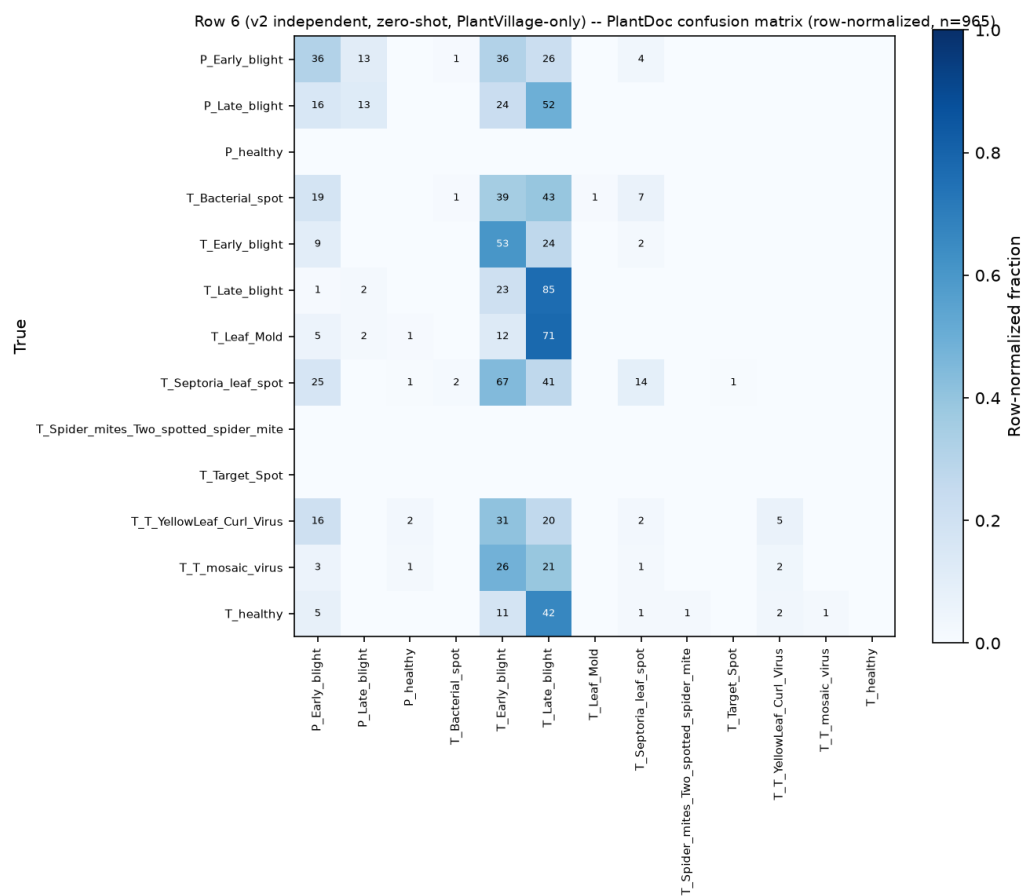
Rows 1, 2, 6 reused directly (already saved, generated 2026-09-16). Row 8 newly generated (docs/report_assets/generate_charts.py, from row8_resnet18_plantdoc_perimage_predictions.json) -- no image existed for it before this report.



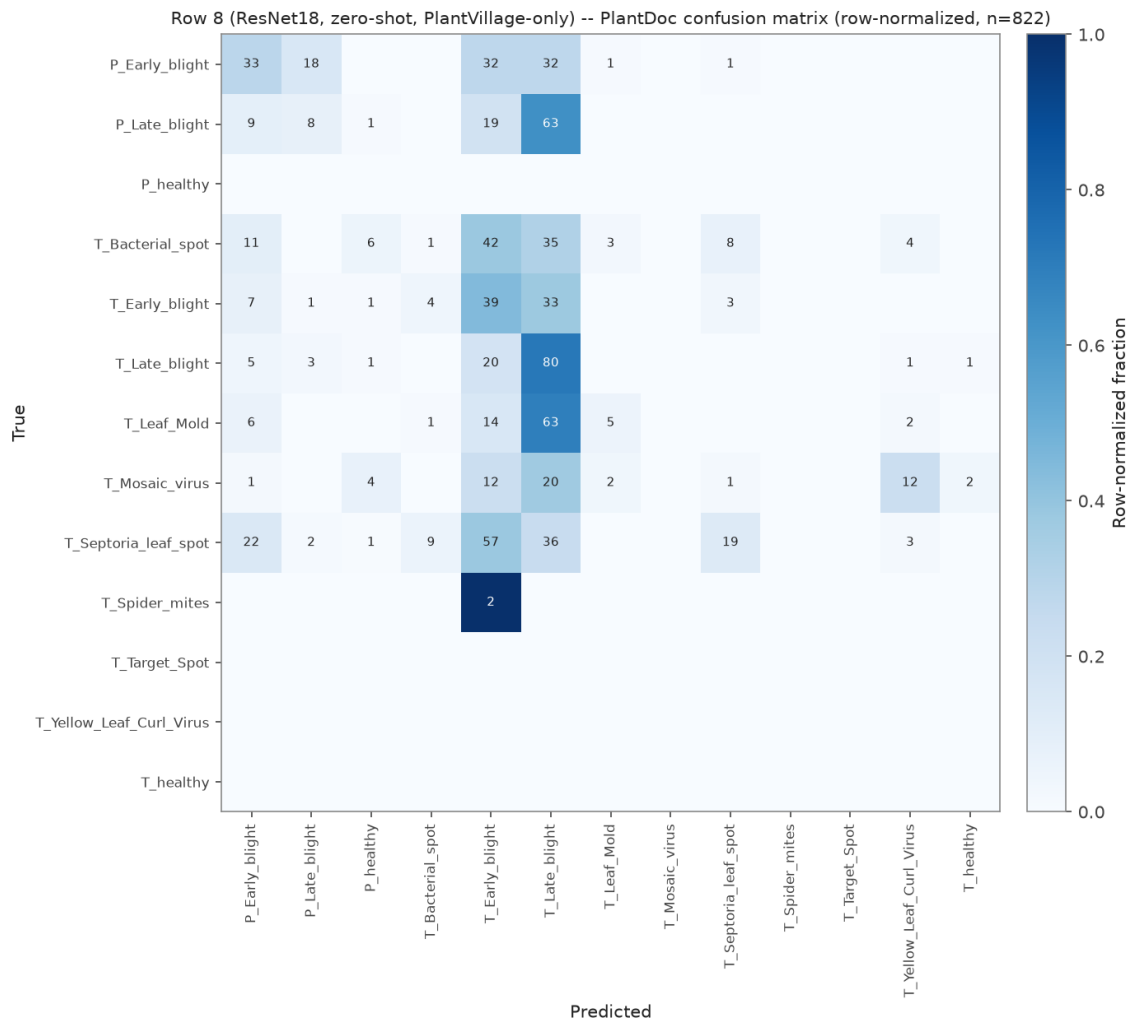
Row 1 (flat baseline, zero-shot) -- reused.



Row 2 (hierarchical, zero-shot) -- reused.



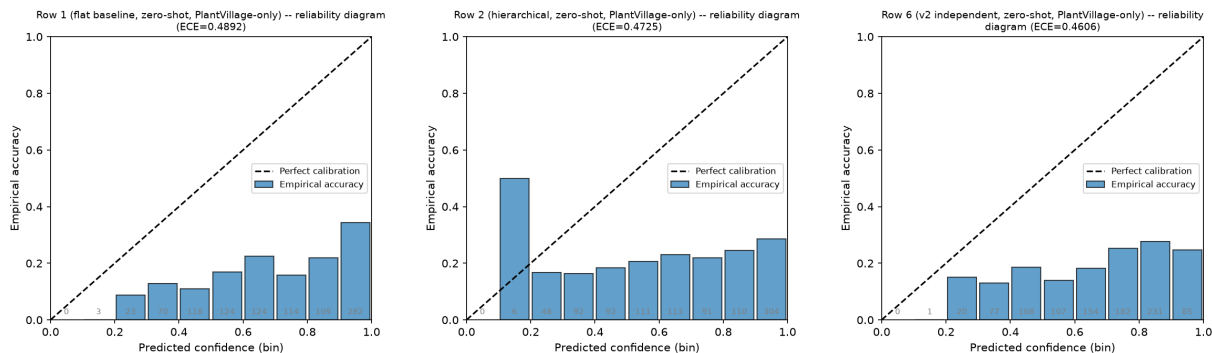
Row 6 (v2 independent checkpoint, zero-shot) -- reused.



Row 8 (ResNet18, zero-shot) -- newly generated for this report.

Reliability diagrams (calibration), rows 1, 2, 6

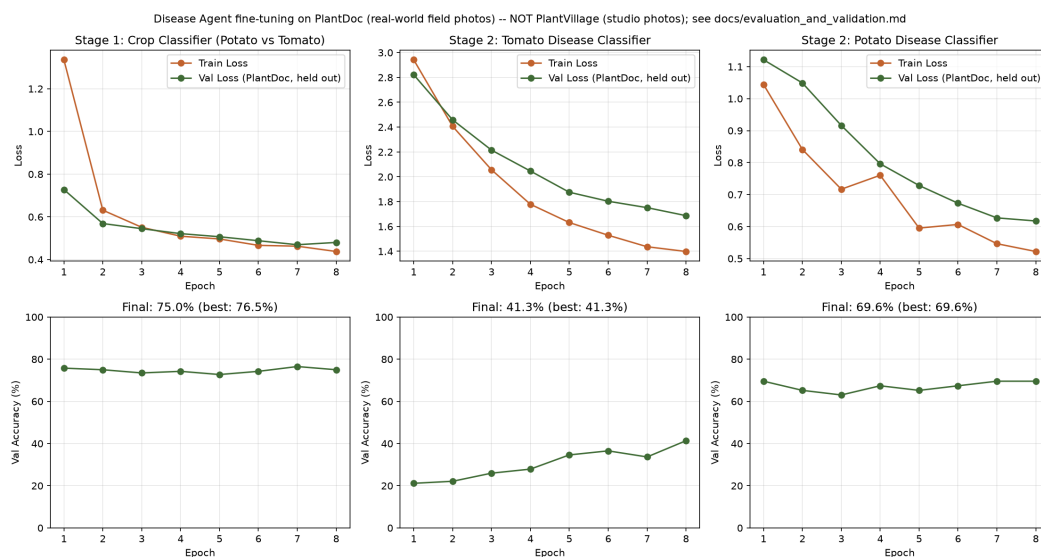
All reused directly from agents/disease/results/ (generated 2026-09-16). Confidence bins above ~0.5 sit well below the diagonal in all three -- the model is systematically overconfident, not just inaccurate.



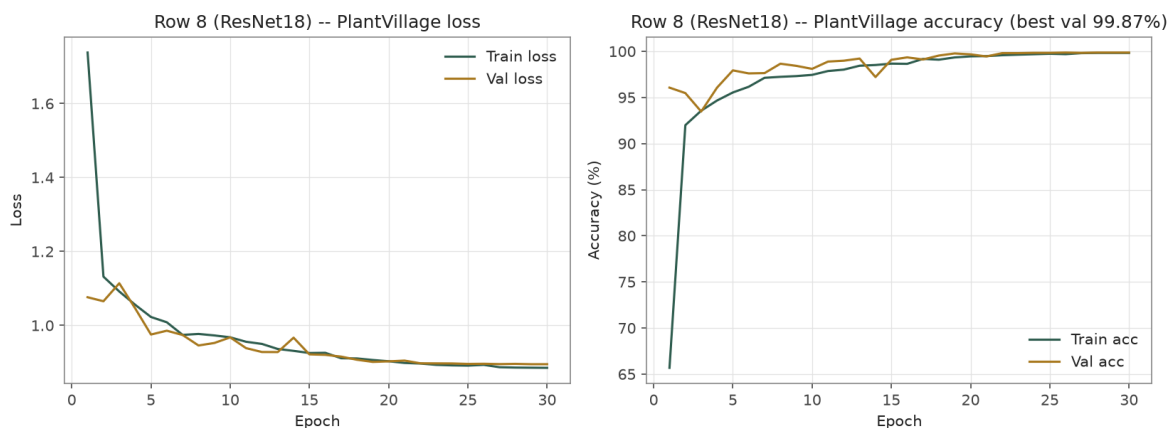
Rows 1 / 2 / 6, left to right -- all reused. ECE: 0.4892 / 0.4725 / 0.4606.

Training curves

Row 5 (deployed model's PlantDoc fine-tune): reused, plantdoc_finetuning_curves.png (generated by agents/disease/plot_finetuning_curves.py). Rows 1, 2, and 6 have **no saved per-epoch training curve** -- only final-epoch summary metrics were recorded for those runs; this gap is stated explicitly rather than silently omitted. Row 8 (ResNet18) has a full 30-epoch history on file and is newly plotted below.

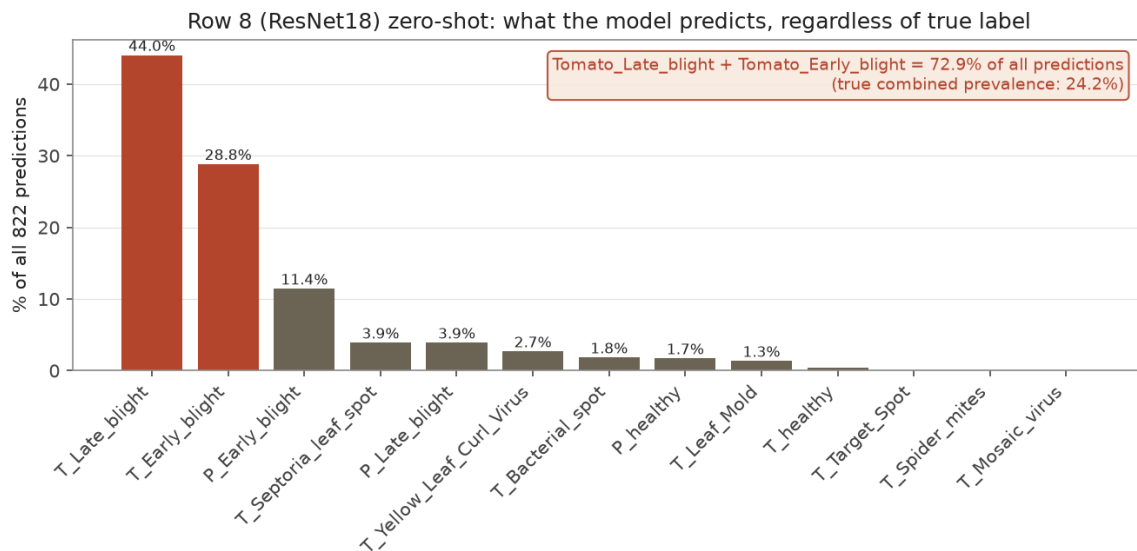


Row 5's PlantDoc validation-accuracy fine-tuning curve -- reused.



Row 8 (ResNet18) PlantVillage loss/accuracy per epoch -- newly generated from docs/evidence/row8_resnet18_training_evidence.json's saved history array.

Attractor-class bias, visualized



Row 8 (ResNet18) prediction distribution -- newly generated from row8_resnet18_plantdoc_zeroshot_evidence.json's prediction_distribution field. The same Tomato_Late_blight/Tomato_Early_blight dominance pattern is documented for rows 1, 2, and 6 in metric1_confusion_matrix_calibration_evidence.json.

Dataset sizes

PlantVillage split sizes (13-class, 2-crop subset)

Checkpoint(s)	Train	Val	Test	Note
Row 1 / 2 (Phase 1 original)	14,535	3,628	n/a (val used as held-out check)	8-epoch AdamW run
Row 8 (ResNet18)	14,218	3,047	3,047	Explicit 70/15/15 split, seed=42

PlantDoc per-class counts (rows 1/2 vs. row 6 vs. row 8)

Class	Rows 1/2 (n=967)	Row 6 (n=965)	Row 8 (n=822)
Potato_Early_blight	116	116	117
Potato_Late_blight	105	105	100
Tomato_Bacterial_spot	110	110	110
Tomato_Early_blight	88	88	88
Tomato_Late_blight	111	111	111
Tomato_Leaf_Mold	91	91	91
Tomato_Septoria_leaf_spot	151	151	149
Tomato_Spider_mites	2	0 (missing)	2
Tomato_Mosaic_virus	54	54	54
Tomato_Yellow_Leaf_Curl_Virus	76	76	0 (missing)
Tomato_healthy	63	63	0 (missing)

Coverage-gap note: Row 8's 822-image evaluation set is missing Tomato_Yellow_Leaf_Curl_Virus and Tomato_healthy entirely (139 images), plus 7 stray misses elsewhere -- 146 images total that rows 1/2's 967-image set has and row 8 doesn't. Most likely an external PlantDoc-Dataset clone/version difference, not independently root-caused. Row 6 is missing exactly the 2 Tomato_Spider_mites images (a separately investigated, negligible ~0.2pp effect). Full detail: model_lineage.md's Row 8 section.

Five-signal convergence

Signal	Finding
1. Accuracy band	22.00% / 23.45% / 21.45% / 22.51% (rows 1/2/6/8) -- mean 22.30% (rows 1/2/6), stdev 1.03pp
2. Attractor-class (confusion matrix)	All 4 checkpoints independently default to Tomato_Late_blight/Tomato_Early_blight regardless of true label
3. Calibration shape (ECE)	0.4892 / 0.4725 / 0.4606 (rows 1/2/6) -- all systematically overconfident
4. Formal statistical confirmation (McNemar's, same-family)	p=0.316 / 0.754 / 0.179, all non-significant
5. Cross-architecture-family confirmation (McNemar's, ResNet18 vs. rows 1/2/6)	p=0.369 / 0.156 / 0.083, all non-significant

Read together: not five separate findings, but the same underlying phenomenon viewed at five resolutions -- replicated across four independently-trained models and two genuinely distinct backbone architecture families (EfficientNetB0 and ResNet18). Strong evidence the ~22% real-world ceiling is a structural covariate-shift problem inherent to PlantVillage-only training data, not a fixable modeling error specific to one run or one architecture.

Per-class recall (row 5, deployed, current checkpoint)

Class	Recall	n (test)
Tomato_Yellow_Leaf_Curl_Virus	66.7%	6
Tomato_Early_blight	55.6%	9
Tomato_Septoria_leaf_spot	54.5%	11
Potato_Late_blight	37.5%	8
Tomato_healthy	37.5%	8
Tomato_Leaf_Mold	33.3%	6
Tomato_Mosaic_virus	30.0%	10
Potato_Early_blight	25.0%	8
Tomato_Bacterial_spot	22.2%	9
Tomato_Late_blight	20.0%	10

n=6-11 per class (sums to 85, matching the 37.65% headline exactly) -- each individual test image is worth roughly 9-17 percentage points of that class's own recall. Source: agents/disease/results/full_verification_metrics.json (the same file backing the precision/recall/F1/latency/footprint table above). Corrected 2026-09-17: an earlier version of this table showed row 7's bug-confirmed-invalid n=71 data (see the model lineage table's row 7 entry) mislabeled as row 5 -- caught by inspecting that the n-values summed to 71, not 85.

Zero-shot vs. fine-tuned: the recorded decision

Confirmed with the project guide, 2026-09-13. The zero-shot PlantDoc result (row 2: 23.45% / 28.24% test-only) is the headline real-world accuracy number for the Disease Agent. The PlantDoc-fine-tuned result (rows 3-5, including the currently-deployed row 5 at 37.65%) is demoted to a secondary, clearly-labeled *exploratory* result -- not presented as a comparison against the zero-shot number, since row 5 was fine-tuned on PlantDoc and is methodologically incomparable to a zero-shot claim.

Bugs found and fixed during this evaluation

Bug / gap	Status	Where documented
Conflict Resolver rule-table gap: (None, rain_risk, sell_now) had no entry	FIXED (Phase 8.1 metric #5)	metric5_conflict_rule_table_evidence.json
Row 7 fine-tune: data-loading bug silently dropped 2 of 13 classes from test set	Found, NOT fixed -- run marked INVALID for headline reporting	model_lineage.md "Row 7" section
Row 6: 2-image PlantDoc discrepancy vs. rows 1/2's 967-image set	Investigated -- effect ~0.2pp, not material, not fixed (traced to local file-count change)	metric1_v2_checkpoint_evidence.json
Row 8 / k-fold CV: PlantDoc-Dataset clone missing 2 classes (139+ images)	Found, documented, not independently root-caused (external Colab environment)	model_lineage.md "Row 8" section
ASR transliterates "pH" -> flips Intent Router classification	Found, confirmed structural (4/4 speakers) -- not fixed, flagged for future work	metric4_voice_vs_text_ablation_evidence.json
TTS 5-7x slower inside Streamlit	Root cause unresolved after 8 hypotheses ruled out -- mitigated by migrating to FastAPI	claude_phase6-summary.md Sec 6.8
Memory collision: Intent Router LLM + TTS on 8GB RAM caused multi-minute stalls	FIXED	claude_phase6-summary.md
Intent Router 87.5% full-set number includes 5 few-shot-exposed questions	Corrected -- 81.8% held-out is the honest estimate	metric_intent_router_per_intent_evidence.json

Algorithm and Methodology

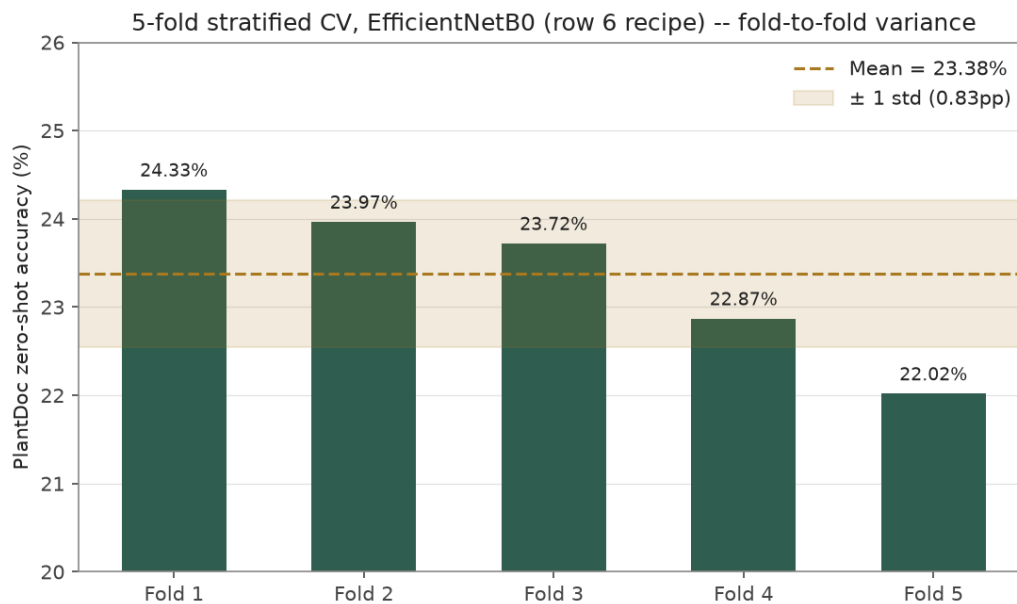
Model	Architecture	Optimizer / LR	Loss	Batch	Epochs / early-stop	Params / FLOPs
Row 1 (flat)	EfficientNetB0 (ImageNet pretrained)	AdamW, lr=1e-4	CrossEntropy, class-weighted	32	8 / none	4.02M / 0.82G
Row 2 (hierarchical)	3x EfficientNetB0 (stage1 crop-ID + 2x stage2 disease)	AdamW, lr=1e-4	CrossEntropy, class-weighted	32	8 / none	12.04M / 1.64G per inference
Rows 3-5 (PlantDoc fine-tune)	Row 2 arch., frozen backbone + last block unfrozen	Adam, head lr=3e-4 / last-block lr=3e-5	CrossEntropy, class-weighted	16	8 / none	same as row 2/5
Row 6 (v2, independent)	EfficientNetB0 (flat)	External Colab run -- optimizer/LR not captured in repo evidence	not captured	not captured	not captured	4.02M / 0.82G
Row 7 (extended fine-tune)	Row 6 arch. + PlantDoc train	base_lr=1e-5	CrossEntropy, class-weighted, label smoothing 0.1	not captured	25 / patience 8	same as row 6
Row 8 (ResNet18)	ResNet18 (ImageNet pretrained)	base_lr=3e-4, seed=42	class-weighted, label smoothing 0.1	32	30 / patience 6	11.18M / 3.65G
Intent Router	qwen2.5:7b-instruct (Ollama, local)	n/a (inference only)	n/a	n/a	n/a	7B params

Intent Router model choice: qwen2.5:3b-instruct scored 50.0% on the 16-question smoke test vs. 7b-instruct's 87.5% (81.8% held-out) -- the 3B model's accuracy drop was judged not worth the latency gain, so 7B was deployed despite its ~12s routing latency cost.

Row 6's optimizer/LR/batch/epoch settings were not captured anywhere in this repository's evidence files (it was trained in an external Colab session with only the final checkpoint and zero-shot results preserved) -- stated as a gap rather than guessed.

K-Fold Cross-Validation

5-fold stratified CV on EfficientNetB0 (row 6's recipe), CV pool = train+val only (original held-out test set untouched throughout). Answers a different question than the multi-checkpoint convergence above: not "do independent models agree," but "if you retrain the same recipe on a different data partition, how much does the number move."



Newly generated from docs/evidence/kfold_cv_row6_variance_evidence.json.

Fold	PlantVillage val acc	PlantDoc zero-shot acc
1	99.86%	24.33%
2	99.88%	23.97%
3	99.80%	23.72%
4	99.86%	22.87%
5	99.80%	22.02%
Mean +/- std	99.84% +/- 0.03%	23.38% +/- 0.83%

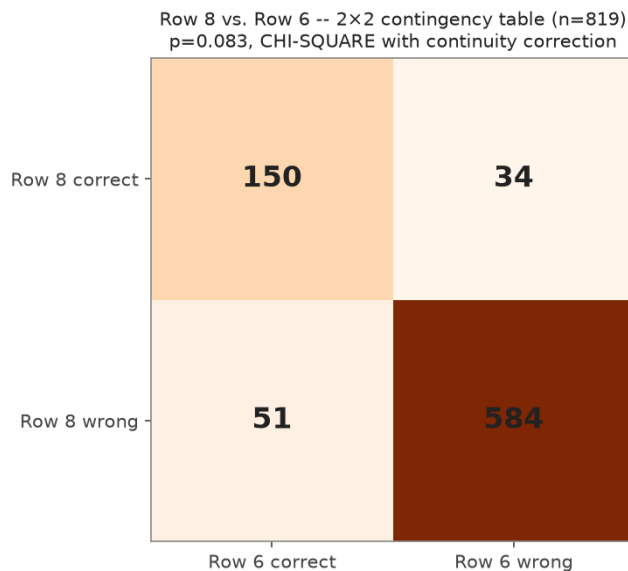
A strictly monotonic downward drift across the 5 folds (24.33% -> 22.02%) is visible in the raw data -- reported as an observed pattern, not claimed as a trend at n=5. Scoped honestly: one architecture only (EfficientNetB0), reduced 15-epoch/fold budget (vs. the original 30), and the PlantDoc image set (n=822 per fold) most likely carries the same coverage gap as row 8's rather than matching rows 1/2/6's fuller 967/968.

McNemar's Test -- Statistical Significance

Tests whether the accuracy-band convergence above is a descriptive observation ("these numbers are close") or genuine statistical support ("cannot reject the null hypothesis that these checkpoints have the same accuracy"). Image-set identity verified via set operations before every test, not assumed.

Pair	n (shared)	Discordant pairs	Statistic	p-value	Significant?
Row 1 vs Row 2	967	168	1.006	0.316	No
Row 1 vs Row 6	965	163	0.098	0.754	No
Row 2 vs Row 6	965	179	1.810	0.179	No
Row 8 vs Row 1	821	150	0.807	0.369	No
Row 8 vs Row 2	821	161	2.012	0.156	No
Row 8 vs Row 6	819	85	3.012	0.083	No

All 6 pairwise comparisons across this project are non-significant at $\alpha=0.05$ (not corrected for multiple comparisons -- noted as a caveat in the underlying evidence files). A 2x2 contingency table visualization for the pair closest to significance (Row 8 vs. Row 6, $p=0.083$):



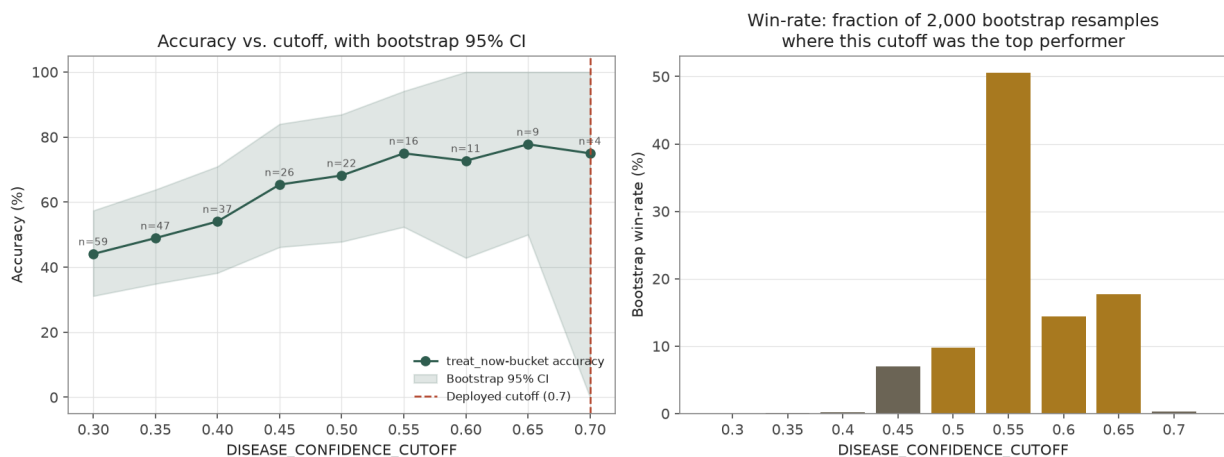
Newly generated from `mcnemar_row8_resnet18_vs_zeroshot_evidence.json`'s raw contingency table.

Metric 5 -- Conflict Resolver

Dataset	12 synthetic scenarios (single-, dual-, triple-agent conflict/agreement)
Result	13/13 correct, 100% (13th = regression test for the live-data-found rule-table gap, now fixed)
Coverage, denominator-honest	7/36 of the full 3-agent-active state grid (3 disease x 4 weather x 3 price)
Limitation	Validates decision logic over synthetic inputs, not live-agent real-world scenario coverage

Metric 6 -- Confidence Calibration / Cutoff Sweep

Sweeping DISEASE_CONFIDENCE_CUTOFF from 0.30 to 0.70 against the deployed checkpoint's 85-image PlantDoc test-only set found several candidate values with higher treat_now-bucket accuracy than the deployed 0.7 -- but on far smaller buckets. A Wilson 95% CI overlap check plus a 2,000-iteration bootstrap resample showed this is a real but imprecise effect.

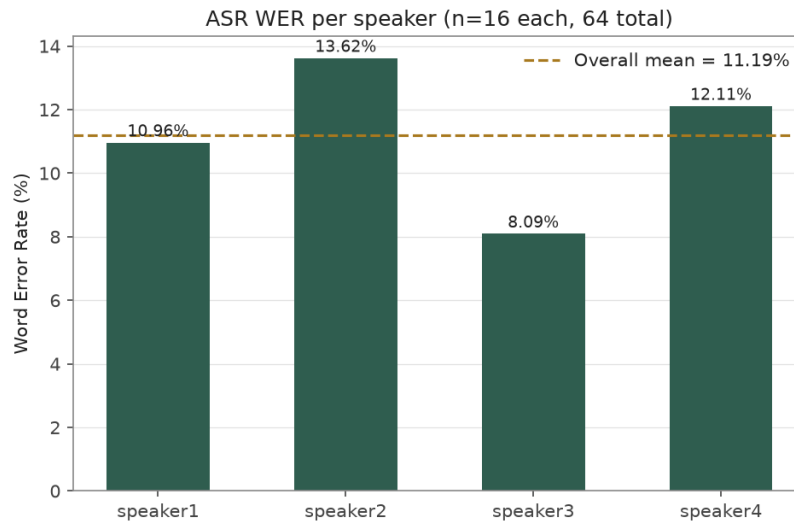


Newly generated from metric6_confidence_calibration_evidence.json's cutoff_sweep_and_stability_check_2026-09-13 data -- no chart image existed for this analysis before this report.

Decision: DISEASE_CONFIDENCE_CUTOFF stays at 0.7 -- no code change. The 0.50-0.65 region wins 92.5% of bootstrap resamples collectively, but no single value can be pinned down as the optimum on this sample size. Reasoning: the deployment risk of changing live pipeline behavior outweighs a marginal, statistically imprecise accuracy gain.

Metrics 2-4 -- ASR, Latency, Voice-vs-Text Ablation

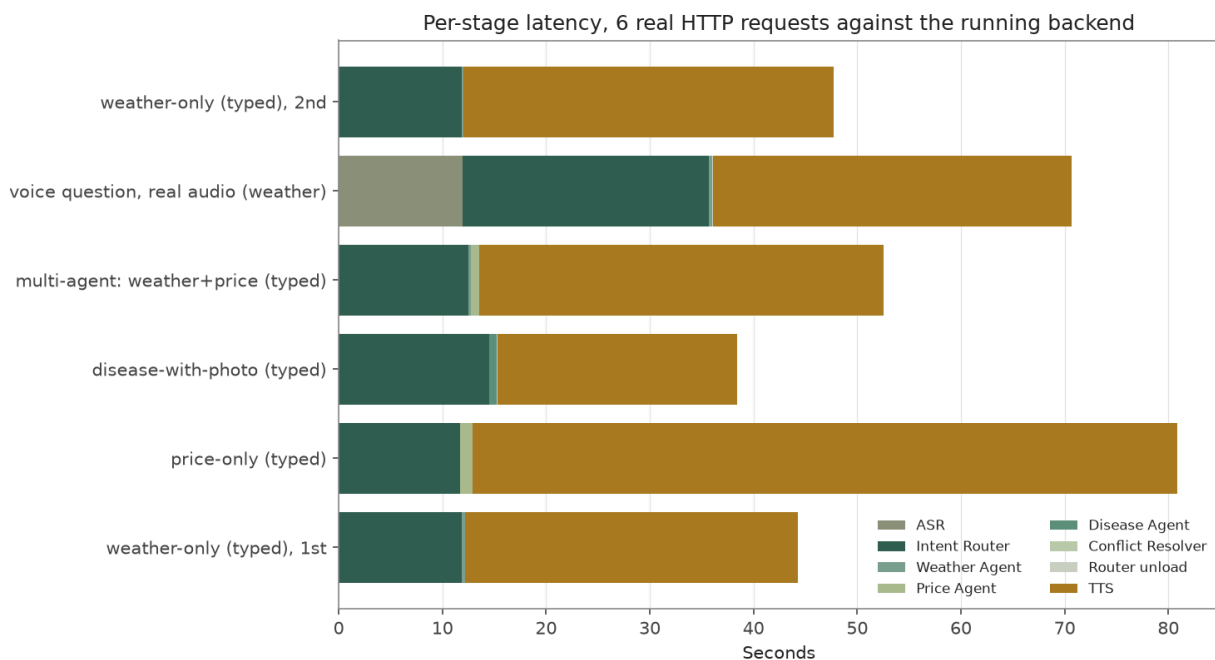
Metric 2 -- ASR Word Error Rate (final, 4/4 speakers)



Newly generated from tests/audio/wer_eval/wer_results.json -- no chart image existed for the per-speaker breakdown before this report.

Overall mean WER 11.19% across 64 utterances (4 speakers x 16 questions). All 4 speakers land within a ~5.5-point band (8.09-13.62%) -- no wild outlier.

Metric 3 -- Per-Stage Latency



Newly generated from metric3_latency_evidence.json's 6 real HTTP-request trials -- no chart image existed for this breakdown before this report.

Intent Router (~12-15s) and TTS (~23-68s) dominate total request latency; the Weather/Price/Disease agents themselves each contribute well under 2 seconds.

Metric 4 -- Voice-vs-Text Ablation

n=64 (16 questions x 4 speakers). Intent-router and agents-fired output matched 60/64; final answer matched 58/64. Of the 6 apparent mismatches, all are now fully explained: 4 are the confirmed structural ASR-transliteration finding below (one per speaker), and 2 were investigated and ruled out as live-API timing differences unrelated to voice-vs-text pipeline logic.

Headline finding: ASR transliterates "pH" and flips the Intent Router's classification. Question 8 ("soil pH for tomato") -- every one of the 4 speakers' independent recordings had ASR transcribe "pH" as its Telugu transliteration instead of the reference text's Latin-script "pH," and all 4 triggered the same intent-router flip from wants_soil=True to wants_price=True. This is a STRUCTURAL finding about the voice pipeline, not an ASR accuracy defect -- domain-specific technical terms crossing the ASR-to-LLM boundary can silently change system behavior in ways a text-only test suite would never catch.

Intent Router: 87.5% -> 81.8% Honesty Correction

14/16 correct (87.5%) on the full 16-question set, matching the original measurement exactly. But 5 of these 16 questions are verbatim few-shot examples inside the router's own SYSTEM_PROMPT. Recomputing on only the 11 genuinely held-out questions gives **9/11, 81.8%** -- the more honest real-world estimate. Both of the full set's 2 failures fall inside the 11 held-out questions, so removing the few-shot-exposed subset concentrates the same 2 errors onto a smaller base rather than diluting them.

Intent	Accuracy	Precision	Recall	F1	n positive
wants_disease	93.8%	85.7%	100%	0.923	6
wants_weather	93.8%	100%	83.3%	0.909	6
wants_price	93.8%	100%	80.0%	0.889	5
wants_soil	100%	100%	100%	1.000	3

Verdict: SPREAD, not concentrated -- the 2 failures touch 3 different intent labels (wants_soil is perfect), so this isn't one weak intent category dragging the average down. The more informative axis is compound- vs. single-intent questions: the only question expecting all 3 intents simultaneously is exactly the one the router does worst on.

Compute Footprint

Row	Params	FLOPs / inference	Latency	Notes
1 / 6 (flat EfficientNetB0)	4.02M	0.82 GFLOPs	not measured	identical by architecture construction
2 / 5 (hierarchical, 3-model)	12.04M (all 3 loaded)	1.64 GFLOPs (2-of-3 per real call)	240.2ms mean (std 20.7ms, n=85), 4.16 img/s -- row 5 only, real measurement	row 5 is the only row with a REAL latency measurement
8 (ResNet18)	11.18M	3.65 GFLOPs	not measured	more FLOPs than EfficientNetB0 despite fewer params -- expected, EfficientNet targets FLOP-efficiency specifically

Params/FLOPs computed via ptflops on the untrained architecture (depends on architecture and class count, not trained weights).
 Row 6/7 and row 8's single-model architectures have never been latency-profiled -- stated honestly as a gap, not estimated from FLOPs alone.

Honestly Documented Limitations

- Disease Agent real-world accuracy is structurally bounded (~21-23% zero-shot) by a covariate-shift domain gap between PlantVillage's lab photos and PlantDoc's field photos -- confirmed architecture-independent, not a fixable modeling error. The deployed fine-tuned model (37.65%) should not be presented as reliable for field deployment.
- Statistical power remains modest on several evaluations: Intent Router (n=16, 11 held-out), Conflict Resolver (n=13 synthetic), ASR (n=64, still a modest speaker count for broad dialect robustness), disease-with-photo end-to-end (n=30).
- Row 6's training hyperparameters (optimizer, LR, batch size, epoch count) were never captured in this repository -- trained in an external Colab session, only the final checkpoint and zero-shot results were preserved.
- TTS latency inside the original Streamlit architecture remains unresolved at the root-cause level (8 hypotheses ruled out, macOS SIP blocked deeper profiling) -- mitigated by migrating to FastAPI, not fixed at the mechanism level.
- The ASR pH-transliteration failure mode is confirmed structural but not fixed -- flagged as a concrete direction for future work (normalizing known technical terms before intent routing), not applied as a fix in this evaluation.
- Row 8 (ResNet18) and the k-fold CV run's PlantDoc evaluation sets carry an unresolved, not-independently-root-caused coverage gap (2 of 13 classes, 139+ images) relative to rows 1/2/6's fuller benchmark -- most likely an external dataset-clone/version difference.
- McNemar's test results (6 pairwise comparisons) are not corrected for multiple comparisons -- a Bonferroni-adjusted threshold ($\alpha=0.05/6=0.0083$) would be more conservative than the $\alpha=0.05$ used throughout.

Conclusion

Krishi-Agent demonstrates that a voice-first, conflict-aware multi-agent farm advisory system can be built with deterministic, independently-tested arbitration logic; empirically-validated (not merely asserted) confidence calibration, now confirmed stable across two architecture families and under data-partition resampling; and a rigorously characterized, honestly-reported set of failure modes at each pipeline boundary -- rather than a single headline accuracy number standing in for the whole system's readiness.

The project's most scientifically interesting result is not a success metric: it is the structural, architecture-independent characterization of the PlantVillage-to-PlantDoc domain gap, formally tested and now reproduced across two distinct backbone families and five independent training runs. Where the system falls short of production readiness -- the Disease Agent's real-world ceiling, the ASR pH-transliteration failure mode, unresolved TTS latency inside the original architecture -- those limits are stated plainly in Section 13, each traced to a specific evidence file, not smoothed over.

Every figure in this report was cross-checked against its backing script or JSON evidence file at generation time; no number was estimated, recreated from memory, or fabricated.